NOVACRM · CUSTOMER SUCCESS

Customer Onboarding Automation

Requirements Analysis

CLIENT
NovaCRM, Customer Success

PREPARED FOR
Rocketlane: Forward Deployed Engineer assessment, Part 1

PREPARED BY
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STATUS
Submitted. Open questions answered by Karthik Ramaiah (Rocketlane, acting for NovaCRM) on 29 Oct 2026; final acceptance pending client UAT (§12)

1. Executive summary

Problem. NovaCRM's Customer Success team onboards every customer by hand, starting from a free-text email the Account Executive (AE) sends when a deal closes. Three failures recur: the **wrong template** is applied because the email doesn't state the plan tier; **data migration is marked done** without the customer verifying it; and **stalled work isn't escalated**.

Proposed solution. An agentic workflow that turns a deal email into a running onboarding project. An Intake Agent extracts and validates the deal details and asks the AE for anything missing; a voice agent phones the AE to confirm the tier; the system creates the Rocketlane project from the matching template and assigns the CSM; a Communication Agent opens a personalised Slack channel. Rocketlane automations escalate overdue tasks and request the customer's migration sign-off. Deterministic code controls every step; language models only read and write text. When anything is missing or uncertain, the system asks a person instead of guessing.

What was built. A working system, verified against live Gmail, a real phone call, Rocketlane and Slack, with 449 automated tests (§10).

Main gaps before production (§11–§12): blocked-task escalation (Priya's original rule) is not yet covered; the migration approval does not yet *block* the phase from closing; and the human escalation console is an API, not a business-user interface.

2. Context, objectives and success measures

NovaCRM (~200 customers, a few new deals a week) runs onboarding in Asana with ~15 tasks across four phases (Kickoff, Data Migration, Configuration, Go-Live). Enterprise customers get a dedicated CSM and a 30-day plan; Growth customers get a pooled CSM and 14 days.

ID	Objective	Measure (proposed target, to be agreed with the client)
BO-1	No template errors	Template matches the AE-confirmed tier on 100% of projects; 0 projects created on an unconfirmed tier
BO-2	Migrated data is verified by the customer	Every migration has a recorded customer approval; migration-related customer escalations trend to 0
BO-3	Slipping work is escalated automatically	100% of tasks overdue ≥ 1 day notify the Project Manager; ≥ 4 days notify the Project Owner
BO-4	No manual set-up	Deal email → project and channel ready in under 1 business hour, excluding AE response time; share of deals needing a person tracked monthly

Non-goals (reasonable goals deliberately excluded from this phase): kickoff scheduling, the end-of-onboarding handoff summary, Salesforce integration, migrating existing Asana projects (§13).

3. Stakeholders

Stakeholder	Role	Needs from the new process
CS Manager (Priya)	Owns onboarding quality; Project Owner; resolves escalations	Correct templates; escalations that don't rely on memory; a record of every decision; a named backup when unavailable
CSM	Runs each onboarding; Project Manager	No manual set-up; early warning on slipping tasks; a way to reassign work
Account Executive	Hands the deal over	One email and one short call; asked only for what is genuinely missing
Customer	Completes customer-side tasks, verifies data	A shared channel on day one; a clear sign-off step for their data
Sales leadership / RevOps	Owns the AE handoff, Salesforce, the AE directory	Minimal AE disruption; ownership of the AE contact list
IT / Security	Approves access and data handling	Least-privilege access; call-recording and AI-data policies (§9)
Rocketlane & Slack admins	Maintain templates, automations, the Slack workspace	Clear configuration ownership after hand-over

4. Current workflow and root causes

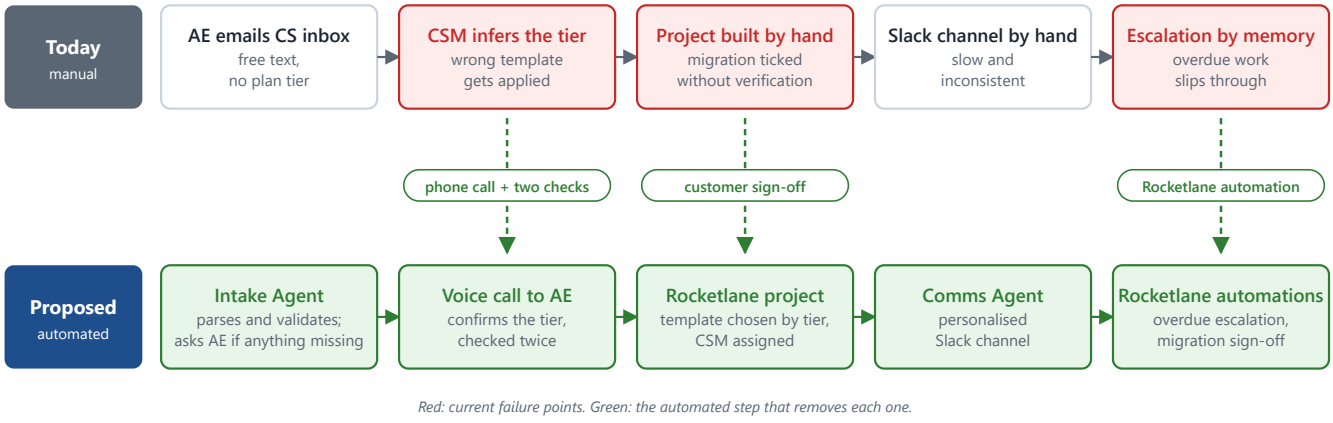


Figure 1. Today's onboarding process, its failure points, and the proposed step that removes each

#	Step today	Owner	Tool	Problem observed
1	Email CS about the closed deal	AE	Email	Free text; plan tier never included; fields sometimes missing
2	Work out the tier and pick a template	CSM	Judgement	Enterprise and Growth templates get mixed up
3	Create the project; work phase checklists	CSM	Asana	Migration tasks ticked without the customer verifying data
4	Create the shared Slack channel; schedule kickoff	CSM	Slack, calendar	Repetitive and inconsistent
5	Escalate tasks blocked > 3 days	CSM	Memory	"Falls through the cracks"
6	Write handoff summary; move to support	CSM	Document	Manual write-up

Problem	Root cause
Wrong template	The tier isn't in the handoff, so a person infers it; nothing checks the choice
Unverified migration	Completion is one click; there is no customer confirmation step
Missed escalations	No system watches task status or due dates
Slow, inconsistent set-up	Data is copied by hand from an unstructured email into several tools

5. Requirements

Priority uses MoSCoW (**M**ust / **S**hould / **C**ould / **W**on't this phase). Each requirement is verified by the test or evidence listed in §10.

5.1 Functional requirements

ID	P	The system shall…	Acceptance criterion
FR-1	M	Monitor the CS Gmail inbox and start one workflow per deal email from an approved AE	Approved sender → exactly one workflow; other senders and non-deal emails logged and ignored
FR-2	M	Extract customer name, contact name and email, AE name and Salesforce link	Every value appears verbatim in the AE's email; nothing inferred
FR-3	M	Validate required fields before proceeding	Missing or malformed field (invalid email, non-Salesforce link) stops the workflow
FR-4	M	Ask the AE for missing information instead of guessing	Reply on the thread lists exactly what's needed; reminder after 24 h; escalation after 48 h
FR-5	M	Phone the AE with a voice AI agent to confirm Enterprise or Growth	AE called on their directory number; tier asked, read back with the customer name, explicit "yes" required
FR-6	M	Create a project only on a clear, unambiguous confirmation	Unclear, missing or contradictory answer never creates a project; no default tier exists
FR-7	M	Retry or escalate when the call fails or is unclear	Retried within business hours, max 3 attempts; then escalated to a person
FR-8	M	Create the Rocketlane project via API from the tier's template and timeline	Enterprise → 30-day template, dedicated CSM; Growth → 14-day template, pooled CSM
FR-9	M	Prevent duplicate onboarding projects	Active project for the same customer → no second project; deal held for a person
FR-10	M	Create a Slack channel shared with the customer, personalised by tier	Name, topic and welcome name the customer, tier and CSM; customer invited via Slack Connect
FR-11	M	Escalate overdue tasks in Rocketlane	Overdue 1 day → Project Manager; 4 days → Project Owner
FR-12	M	Obtain the customer's sign-off on migrated data	Customer approval is requested on the verification task. **Gap:** it does not yet block the phase from closing (§12)
FR-13	M	Log every agent action	Timestamp, actor, input, output and rationale recorded per step
FR-14	M	Let a person take over any stalled deal	Escalation reaches the CS Manager by email and Slack with context; they can confirm the tier, retry, re-read the email or close the deal; their decision is recorded
FR-15	S	Escalate tasks *blocked* for more than 3 business days	**Not built this phase** (client confirmed overdue-only, Q5); Rocketlane status automation planned (§13)

5.2 Non-functional requirements

ID	Area	Requirement
NFR-1	Reliability	No step fails silently: every external call has a time limit, bounded retries and a defined outcome (success or escalation)
NFR-2	Integrity	Retries and restarts never create a second call, project, channel or message for the same deal
NFR-3	Auditability	Any deal's full history, including the reason for each decision, can be reviewed afterwards
NFR-4	Security	Only approved senders start a workflow; webhooks authenticated; email and call content treated as untrusted (prompt-injection resistant)
NFR-5	Calling etiquette	Calls only in business hours, at most 3 per deal
NFR-6	Changeability	Required fields, templates, CSM pools, retry limits and calling hours change by configuration, not code
NFR-7	Scale	~200 customers, a handful of deals per week; production path for higher volume in §11

5.3 Constraints and dependencies

- The tier must be confirmed **by a real voice call**; projects created **through the Rocketlane API**; overdue escalation must be a **Rocketlane automation**, not an agent (from the brief).
- Dependencies: Gmail API (OAuth), Vapi and Twilio telephony, OpenAI, Rocketlane API and automations, Slack (Slack Connect is a paid feature).

6. Clarifications and assumptions

Questions asked (28 Oct, answered 29 Oct by Karthik Ramaiah for NovaCRM). Each answer became configuration, not code.

#	Question	Answer	Resulting decision
Q1	Does the email contain the contact email and AE phone?	Contact email yes; AE phone from a directory	AE directory is the phone source; required fields configurable
Q2	How are CSMs chosen; who is PM and Owner?	Round-robin (Enterprise), pooled (Growth)	CSM = Project Manager; CS Manager = Project Owner
Q3	Same tasks for both tiers?	Yes, durations differ	Two templates, same 15 tasks, 30 vs 14 days
Q4	Confirm the customer on the call? Retries?	Yes; 3 attempts	Read-back includes the customer; then escalate to the CS Manager
Q5	Blocked tasks as well as overdue?	Overdue only	FR-15 deferred and listed as a residual gap
Q6	Customer sign-off on migration?	Yes	Rocketlane approval on the verification task

Assumptions (to be validated with the client)

ID	Assumption	Impact if false
A1	Rocketlane replaces Asana for **new** onboardings; active Asana customers are not migrated	Duplicate check can't see customers still onboarding in Asana: needs a cut-over list
A2	Only Enterprise and Growth exist	New tiers need a template and configuration entry
A3	The tier comes from the AE call, not Salesforce	If the tier is reliable in Salesforce, the call becomes a confirmation only (§13)
A4	AEs work 09:00–19:00 IST, Mon–Fri	Per-AE time zones needed for distributed teams

ID	Assumption	Impact if false
A5	Same company name means same customer	Upsells or re-onboardings could be held wrongly; production should match on Salesforce account ID
A6	RevOps maintains the AE directory and sender allowlist	Stale data → wrong number called or deals ignored

7. Proposed agentic workflow

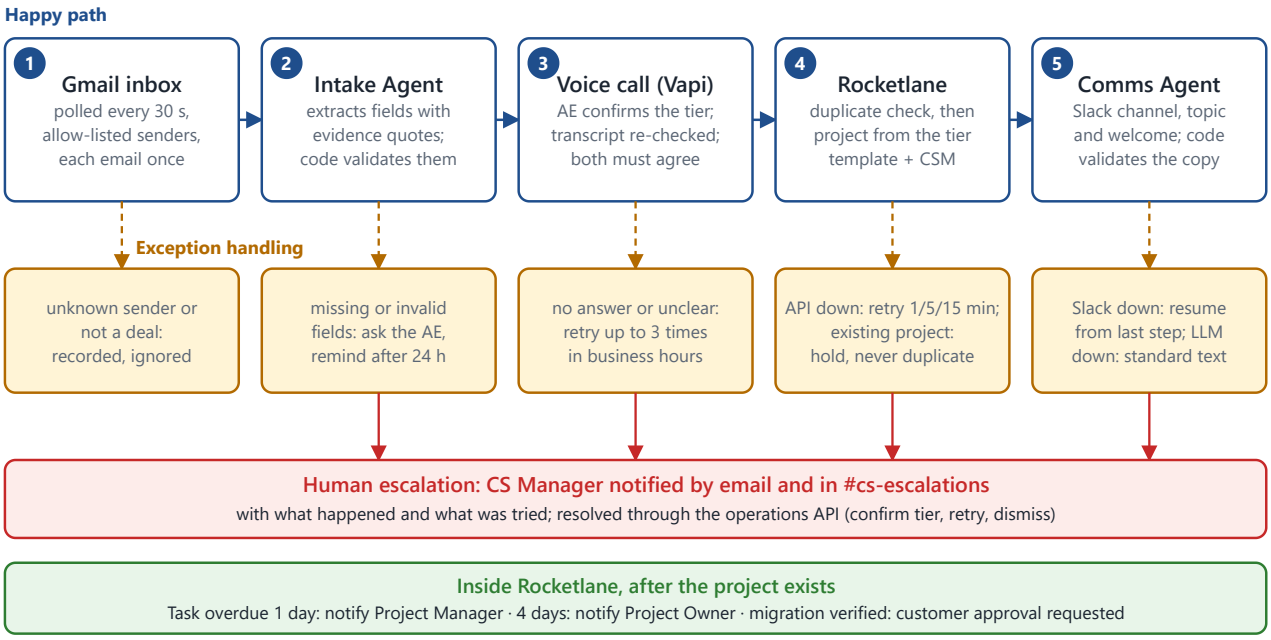


Figure 2. Proposed workflow: the happy path, how each step fails safely, and where a person takes over

Decision rights: who may decide what, and why.

Actor	Kind	Decides	May not	Why this split
Intake Agent	LLM, no tools	What the email says (fields + quotes), whether it's a deal	Create, call or infer the tier	Free text needs language understanding; quotes make every value checkable
Voice agent	LLM voice assistant (Vapi)	How to phrase the conversation; records the AE's answer	Suggest a tier or proceed on its own	A real conversation is required; it only captures the answer
Tier Verifier	LLM, no tools	Independent reading of the transcript	Override the voice record	A second, independent check catches mis-hearing; disagreement = unclear
Workflow engine	Deterministic code	Every action and every business rule	Act on unvalidated data	Rules must be predictable, testable and auditable
Communication Agent	LLM, no tools	Wording of the topic and welcome	Post unchecked text	Personalised tone; code checks it names customer, tier and CSM, else a standard message
Rocketlane automations	Rocketlane configuration	Overdue alerts, approval requests	n/a	Lives where the CSMs work; no custom code to maintain
CS Manager	Person	Anything escalated	n/a	Final authority; every decision logged

Alternatives considered: reading the tier from Salesforce (rejected for this phase; the brief requires a call, and it stays as a future pre-fill); building escalation in code (rejected; Rocketlane automations are maintainable by the client); a single LLM check of the call (rejected; two independent records with a required read-back are stronger against mis-

hearing).

8. Exceptions and guardrails

ID	Situation	Required behaviour
E-1	Missing customer name or contact email	Stop; ask the AE for exactly what's missing; remind at 24 h; escalate at 48 h
E-2	Value not actually in the email	Treated as missing; inferred values are never accepted
E-3	Unknown sender or non-deal email	Logged and ignored; no call
E-4	AE doesn't answer	Retry after 10 and 30 min within business hours; escalate after 3 attempts
E-5	Ambiguous answer ("probably the bigger one")	Asked once more; still unclear → recorded as unclear, retried, then escalated; never mapped to a tier
E-6	Voice record and transcript disagree	Treated as unclear
E-7	Rocketlane unavailable	Retry after 1, 5 and 15 min, then escalate; never assume success
E-8	Request timed out but project was created	Found by its deal reference and adopted; no duplicate
E-9	Customer already has an active project	Held and escalated; no second project
E-10	Slack or the copy model unavailable	Resume from the last completed step; standard message replaces generated copy

9. Security, privacy and compliance

Topic	This build	Required for production
Call recording and consent	Calls are recorded and transcribed by Vapi	Recording disclosure at the start of the call; consent rules per AE jurisdiction; retention policy
Customer data sent to AI	Email text and transcripts go to OpenAI	Enterprise data-processing agreement, zero-retention settings, data-minimised prompts
Data retention	Audit log keeps inputs and outputs indefinitely; secrets masked	Retention period, PII redaction in logs, deletion on request
Access	Operations API behind a single admin token	SSO with role-based access; read-only audit role
Integrations	OAuth for Gmail; API keys in an environment file	Secrets manager, key rotation, least-privilege scopes reviewed by IT
Phone source	AE directory first; a number in the email only if the AE isn't listed	Directory only, so an email can never redirect a call

10. What was built and how it was verified

Built: the full workflow in Python (FastAPI, OpenAI Agents SDK), with Gmail polling, Vapi + Twilio voice, the Rocketlane and Slack APIs, a durable job queue and an audit log. Two Rocketlane templates and both automations are configured in the client workspace.

Live verification (29 Oct 2026): a deal email read and validated from Gmail; a real phone call in which the AE confirmed "Enterprise" and both checks agreed; a Rocketlane project created from the Enterprise template with a dedicated CSM; the personalised Slack channel with an agent-written welcome; a live escalation email and Slack alert

when the intake model failed; a timed-out Rocketlane create adopted without a duplicate. Outage and no-answer paths are verified by automated tests rather than live outages; the overdue automation is configured but has not yet fired on a real task.

Traceability: problem → requirement → guardrail → evidence

Problem	Requirements	Guardrails	Automated test (or evidence)
End-to-end handoff	FR-1–FR-10	—	`test_deal_email_to_onboarding_channel` (email → channel, every job through its real entry point)
Missing or wrong data	FR-2, FR-3, FR-4	E-1, E-2, E-3	`test_missing_required_field`, `test_hallucinated_value_is_rejected`, `test_gaps_ask_the_ae_on_the_thread`, `test_no_reply_gets_a_reminder_then_escalates`, `test_unknown_sender_is_recorded_as_ignored`
Wrong template	FR-5–FR-8	E-4, E-5, E-6	`test_project_is_created_from_the_tiers_template` (both tiers), `test_dedicated_csms_rotate_across_deals`, `test_no_answer_schedules_retry_inside_window`, `test_unclear_answer_is_retried_not_guessed`, `test_keys_disagree_is_unclear`
Duplicates and outages	FR-9, NFR-1, NFR-2	E-7, E-8, E-9, E-10	`test_rocketlane_down_retries_then_escalates`, `test_timed_out_create_is_adopted_not_duplicated`, `test_existing_project_holds_and_escalates`, `test_slack_down_retries_and_resumes_without_duplicates`
Inconsistent set-up	FR-10	E-10	`test_bad_copy_is_rejected`, `test_fallback_is_tier_specific`, `test_two_bad_drafts_fall_back_to_the_template`
Missed escalations / unverified data	FR-11, FR-12	—	Rocketlane configuration (screenshot in the demo)
Auditability and human control	FR-13, FR-14	—	Audit trail per deal; `test_human_tier_confirmation_queues_the_project`

11. Production readiness

Area	This build	Change for production	Priority
Customer in the Slack channel	Real Slack Connect invite (`conversations.inviteShared`) to the customer contact	Track invite acceptance and remind if it is still pending	Should
Escalation console	API calls with an admin token	Slack buttons or a Rocketlane form for the CS Manager	Must
Duplicate detection	Normalised company name	Match on Salesforce account or opportunity ID; name match only as a prompt for review	Must
Telephony	Twilio trial: a trial notice plays and only verified numbers can be called	Paid account, local caller ID per AE country	Must
Privacy and access	§9	§9	Must
Email trigger	Polling every 30 s via a single Gmail account	Gmail push (Pub/Sub); shared mailbox with delegated access	Should
Hosting and data	One process, SQLite, ngrok for webhooks	Managed hosting, PostgreSQL, real ingress with TLS	Should

Area	This build	Change for production	Priority
Operations	Health check, logs, traces	Alerts on escalations and stuck deals, run-book, on-call owner, cost per deal tracked	Should
Time and capacity	IST calling hours; CSM round-robin ignores leave and load	Per-AE time zones; capacity-aware assignment with manual override	Could

12. Risks and residual gaps

Risk / gap	Impact	Mitigation
Blocked tasks (Priya's ">3 days blocked") are not escalated; only overdue tasks are	Stuck-but-not-late work still slips	Rocketlane automation on status *Blocked* (FR-15), first item on the roadmap
Migration approval doesn't gate the phase: requested when the task completes, but the phase can still close	Unverified data can still reach production import	Make production import depend on an approved verification; CSM checklist until then
AE unreachable for days	Onboarding start delayed	Retries in business hours; CS Manager can confirm the tier manually
Model misreads email or transcript	Wrong data or tier	Quoted values, two independent tier checks; anything uncertain goes to a person
CS Manager is a single point of escalation	Escalations wait during absence	Named backup owner; escalations also posted to a shared channel
AE fatigue from calls	Sales pushback	One short call, business hours only; Salesforce pre-fill planned

13. Scope, rollout and roadmap

In scope	Out of scope this phase (reason)
Intake, validation, clarification; voice tier confirmation; Rocketlane project and CSM; personalised Slack channel; overdue escalation and migration approval; audit trail and human resolution; automated tests	Kickoff scheduling (needs calendar access and customer availability); handoff summary to support (needs end-of-project data); blocked-task escalation (Q5); Salesforce (A3); Asana migration (A1)

Rollout: (1) **Pilot:** run alongside the current process for 2–3 weeks on new deals, with the CS Manager reviewing each one. (2) **UAT:** the client runs the edge cases in §8 and signs off against §2's measures. (3) **Cut-over:** new deals go through the system only; CSMs and AEs get a one-page guide. (4) **Ownership:** Rocketlane templates and automations stay with the Rocketlane admin; the AE directory with RevOps; the system with a named engineering owner.

Roadmap: blocked-task escalation → migration approval gating the phase → Salesforce tier pre-fill → kickoff scheduling → handoff summary at Go-Live.

Appendix A. Solution architecture

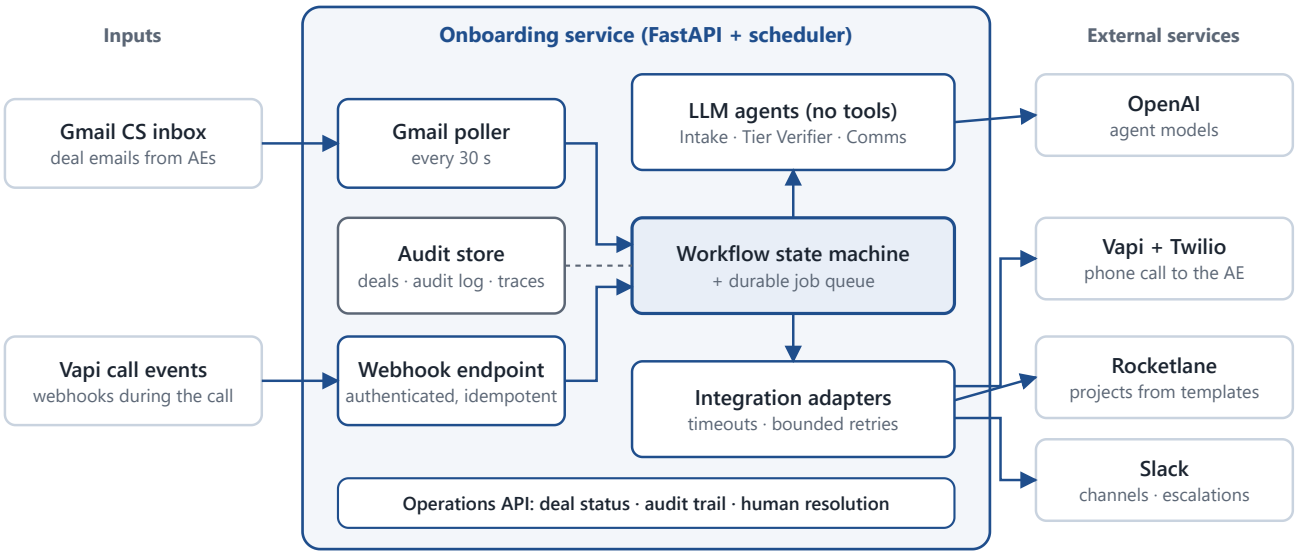


Figure 3. Solution architecture

Technology: Python 3.12, FastAPI, OpenAI Agents SDK, Pydantic, SQLite (PostgreSQL in production), APScheduler, Vapi with Twilio, Rocketlane REST API, Slack Web API.